



Malika Nishnatha

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Profile

A Software Engineering undergraduate with a keen interest in DevOps and how software systems are built, deployed, and maintained. I enjoy learning how things work behind the scenes and am always looking for ways to improve my skills and take on new challenges. I work well in teams and like to contribute in a positive way to whatever I am part of. My goal is to gain hands-on industry experience in DevOps, keep growing as an engineer, and build systems that are reliable and easy to manage.

Education

Bachelor of Science in Computer Science

University of Colombo School of Computing

2024 – present

○ GPA: 3.3/4.0

○ **Coursework:** Data Structures and Program Design in C, Discrete Mathematics, Introduction to Software Engineering, Problem Solving Strategies, Computer Systems, Linear Algebra, Probability and Statistics, Foundations of Algorithms, Database Management Systems, Object Oriented Modeling and Programming, Internet and Web Technologies, Operating System Concepts, Calculus, Software Architectures, Advanced Algorithms, Computer Networks, IT Project Management, Numerical Methods, Artificial Intelligence, Digital Signal Processing, Cryptography and Information Security, Computational Models, Computer System Architecture, Middleware Architecture, Networks and Physical Computing

Bachelor of Science in Computer Science

Sri Lanka Institute of Information Technology (SLIIT)

2023 – 2024

○ Completed one year of the BSc (Hons) in Computer Science programme

○ **Coursework:** Engineering Mathematics, Discrete Mathematics, Programming Methodology, Communication Skills, Computational Thinking

Richmond College, Galle

2009 – 2022

G.C.E. Advanced Level – Physical Science Stream

○ Results: 1 A, 2 B's with 1.68 Z-Score (Combined Maths, Physics, ICT, English)

G.C.E. Ordinary Level

○ Results: 9 A's (English Literature, ICT, Tamil)

Achievements

Virtusa

Champions – CodeQuest CTF Challenge

2025

Part of the winning team in this cybersecurity and capture-the-flag competition organised by Virtusa. Demonstrated technical expertise across CTF challenges and secured first place through strong teamwork and problem solving.

IEEE & ACM Student Branches, University of Colombo School of Computing

Champions – ReidXtreme 4.0

2025

Secured first place in ReidXtreme 4.0, a competitive programming hackathon organised by the IEEE and ACM Student Branches of UCSC. The competition, held on 18 October 2025, served as a training ground for IEEEExtreme – the global 24-hour IEEE programming challenge. Advanced through preliminary rounds and competed in an 8-hour final hackathon, demonstrating strong algorithmic thinking and teamwork under pressure.

Projects

GoPlay – Sports Community Platform (Production)

→ www.goplay.lk

- After receiving an A for the original university project, independently rebuilt and upgraded GoPlay into a production-grade web application now live at www.goplay.lk. Real sports facility owners are currently using the platform to manage and list their facilities, with the system actively serving clients during an ongoing testing period.
- **Tools & Technologies:** Next.js, TypeScript, Tailwind CSS, PostgreSQL, Prisma ORM, Git, GitHub, Vercel
- **Status:** Ongoing – live in production with real clients

GoPlay – Sports Community Platform (University Project)



- Second year group project: a multi-role sports marketplace connecting users with sports facilities, coaches, and equipment retailers. Built entirely from scratch using a custom MVC framework with no external frameworks, and received an A grade upon evaluation by the university supervisor.
- **Tools & Technologies:** PHP, Vanilla JavaScript, custom MVC Architecture, MySQL, HTML, CSS, Git, GitHub, Figma
- **Status:** Done

SwiftTrack – Logistics Middleware Platform (Group Project)



- Designed and implemented a scalable middleware architecture for Swift Logistics, a last-mile delivery company in Sri Lanka. The system integrates three independent legacy systems (CMS, ROS, WMS) into a unified platform supporting real-time tracking and high-volume asynchronous order processing.
- **Tools & Technologies:** Microservices Architecture, RabbitMQ (Publish–Subscribe), Saga Pattern, REST API, SOAP/XML, TCP/IP, Docker
- **Status:** Done

NexusEnroll – Microservices Course Enrollment System (Group Project)



- A university course enrollment management system built using pure Java and microservices architecture. The system consists of three independent microservices (Student, Faculty, Admin) that communicate via REST APIs, implementing five core design patterns – Command, Observer, Strategy, Factory, and Builder – to ensure scalability, maintainability, and robust functionality.
- **Tools & Technologies:** Java, Microservices Architecture, REST APIs, SOLID Principles, UML, Design Patterns (Command, Observer, Factory, Builder, Strategy, Repository)
- **Status:** Done

Technical Proficiency

Languages: JavaScript, TypeScript, PHP, C++, SQL, HTML5, CSS3

Frameworks & Libraries: Node.js, Express.js, React, Next.js, , Tailwind CSS

Databases: MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Docker, Jira, Figma, Vercel

Concepts & Methodologies: RESTful APIs, Microservices, MVC Architecture, Agile, OOP

Extracurricular Activities & Voluntary Work

University of Colombo School of Computing

Colors Awarded – Kabaddi

2025

- Represented the university Kabaddi team in the Inter-University SLUG (Sri Lanka University Games) Tournament 2025. Awarded University Colors in recognition of representing the university at the inter-university level and contributing to the team's performance throughout the tournament.

AIESEC in University of Colombo

Member

2024 – Present

- Active member of AIESEC, the world's largest youth-run organisation, focused on developing leadership skills and global perspectives through community initiatives and cross-cultural collaboration.

Rotaract Club of University of Colombo

Member

2024 – Present

- Participated in community service projects and professional development initiatives, contributing to a spirit of fellowship and giving back to the community.

Others

Ludo Game



- An upgraded version of the classic Ludo game built in C, extended with additional obstacles and challenges beyond the traditional rules. The game runs automatically in the terminal – players are simulated and the game progresses through multiple rounds via a single command until one player is declared the winner.
- **Tools & Technologies:** C
- **Status:** Done

Snake and Ladders



- An enhanced Snake and Ladders game built in C, featuring extra obstacles and challenges beyond the standard game rules. The game runs fully automated in the terminal – triggered by a single command, it simulates all players through multiple rounds and determines the winner automatically without manual input.
- **Tools & Technologies:** C
- **Status:** Done

Arduino Follow Me Robot



- An autonomous path-following robot built on Arduino. The robot uses sensors to detect and follow a path automatically, demonstrating embedded systems programming and hardware integration.
- **Tools & Technologies:** Arduino, C++ (Arduino IDE), Sensors, Embedded Systems
- **Status:** Done

Arduino Automated Plant Nursery



- An IoT-based automated plant nursery system that monitors and controls essential plant growth conditions such as temperature, humidity, soil moisture, and light. All parameters can be monitored and controlled remotely via a mobile application.
- **Tools & Technologies:** Arduino, C++ (Arduino IDE), IoT Sensors, Mobile Integration
- **Status:** Done